Spec → Code: RID010

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1. Spec

REPORT ID

RID010

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report lists subjects with gaps or overlaps in Toxicity Grade Dates, with AE Onset Date not equal to the earliest Toxicity Grade Onset Date OR with AE End Date that is not equal to the latest Toxicity Grade Stop Date.

COLUMN HEADING

1. Country
2. Site
3. Subject Number
4. AE Seq
5. AE Term
6. AE Onset Date
7. AE End Date
8. Tox Grade Onset Date
9. Tox Grade Stop Date
10. Error Message

LOGIC

This report lists subjects with the following conditions:

1) Add entry Toxicity Grade Date overlaps. Output an error message that says "Overlap with Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy"

2) Add entry Toxicity Grade Date gaps. Output an error message that says "Gap with previous Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy"

3) AE Onset Date does not equal the minimum date of Toxicity Grade Onset Date in the add entry section of the same AE. Output an error message that says "AE Onset not earliest Tox Grade Onset"

4a) AE End Date does not equal the maximum date of Toxicity Grade Stop Date in the add entry section of the same AE. Output an error message that says "AE End not latest Tox Grade Stop"

4b) AE End Date is ongoing but there is no blank Toxicity Grade Stop Date in the add entry section of the same AE. Output an error message that says "AE End not latest Tox Grade Stop"

4c) AE End Date has a date but there is a blank Toxicity Grade Stop Date in the add entry section of the same AE. Output an error message that says "AE End not latest Tox Grade Stop"

5) AE Term is completed but Add-entry section is blank, either because it's not initialized or because all the records are completed with UNK/ND/NA. Output an error message that says "Add entry missing"

Report should not display deleted records.

Add-entry records with both dates as UNK/ND/NA are not be considered for scenarios 1 to 4 when there are other records with data entered.

If "Randomized only" parameter selection checkbox is checked, the report should only output subjects with Randomization number that is not null.

REPORT SCENARIOS

Test Case Workbook

SORTING

Country, Site, Subject, AE Seq

2. Logic Understanding

This report cross-checks each AE record's Onset/End dates against a nested add-entry log of Toxicity Grade Onset/Stop date pairs, flagging five categories of discrepancies: overlapping tox grade periods, gaps between tox grade periods, AE Onset not matching earliest tox grade onset, AE End not matching latest tox grade stop (including ongoing/blank mismatches), and entirely missing add-entry data despite a completed AE Term. Deleted records are excluded, UNK/ND/NA-only add-entry records are excluded from date comparisons when real data exists elsewhere for that AE, and an optional 'Randomized only' filter restricts output to subjects with a non-null randomization number. The output lists one row per detected error per AE, sorted by Country, Site, Subject, and AE Seq.
1. For each AE record, first exclude any deleted records entirely -- the report should not display deleted records at any level.

2. Within the add-entry section for a given AE, identify any add-entry records where both Tox Grade Onset Date and Tox Grade Stop Date are UNK/ND/NA; if there exist OTHER add-entry records for the same AE with actual data entered, exclude these UNK/ND/NA-only records from consideration in scenarios 1 through 4.

3. Scenario 1: Examine the remaining (non-excluded) add-entry records for the AE and detect where Toxicity Grade date ranges overlap (i.e., the onset/stop periods of consecutive add-entry records overlap in time). Output an error message formatted as 'Overlap with Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy' identifying the overlapping dates.

4. Scenario 2: Examine the remaining add-entry records and detect where there is a gap between one Tox Grade Stop Date and the next Tox Grade Onset Date (i.e., the periods are not contiguous). Output an error message formatted as 'Gap with previous Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy' identifying the relevant dates.

5. Scenario 3: Compare the AE Onset Date to the minimum (earliest) Tox Grade Onset Date among the add-entry records for that same AE. If they are not equal, output 'AE Onset not earliest Tox Grade Onset'.

6. Scenario 4a: Compare the AE End Date to the maximum (latest) Tox Grade Stop Date among the add-entry records for that same AE. If they are not equal, output 'AE End not latest Tox Grade Stop'.

7. Scenario 4b: If the AE End Date is ongoing (i.e., blank, indicating the AE has not resolved), check whether any add-entry record for that AE has a blank Tox Grade Stop Date. If NO add-entry record has a blank Tox Grade Stop Date, output 'AE End not latest Tox Grade Stop' -- because an ongoing AE should be paired with an ongoing (blank) toxicity grade period.

8. Scenario 4c: If the AE End Date has an actual completed date value, check whether any add-entry record for that AE has a blank Tox Grade Stop Date. If one DOES exist, output 'AE End not latest Tox Grade Stop' -- because a resolved AE should not have an open-ended toxicity grade period.

9. Scenario 5: If the AE Term field is completed (i.e., an AE record exists) but the entire add-entry section is blank -- either because it was never initialized or because all of its records only contain UNK/ND/NA values with no other real data -- output 'Add entry missing' instead of running scenarios 1-4 for that AE.

10. Apply an overall filter: if the 'Randomized only' parameter checkbox is checked at report run time, restrict the entire output to only subjects whose Randomization Number is not null; subjects without a randomization number are excluded regardless of other error conditions found.

11. For each qualifying error found in steps 3-9, output a row containing Country, Site, Subject Number, AE Seq, AE Term, AE Onset Date, AE End Date, Tox Grade Onset Date, Tox Grade Stop Date, and the corresponding Error Message, then sort the final report by Country, Site, Subject, AE Seq. *(inferred)*

3. Spec Review

Readiness: Needs revision — The spec has a solid overall structure but is not ready: the Report Scenarios section is essentially empty ('Test Case Workbook' with no enumerated cases), and the gap/overlap definitions plus several UNK/ND/NA exclusion rules are ambiguous enough to materially change output. Additional consistency/testability concerns around the identical 4a/4b/4c error messages should also be resolved before query generation.

BLOCKING (completeness): Report Scenarios simply says 'Test Case Workbook' with no actual scenarios enumerated in the spec text. There is no way to verify that each Logic condition (1, 2, 3, 4a, 4b, 4c, 5) has a corresponding scenario, or that the scenarios are distinguishable from one another. → Enumerate the actual test scenarios inline (or attach and reference specific scenario IDs/rows from the workbook) so each Logic branch (overlap, gap, onset mismatch, end mismatch variants, missing add-entry) has a traceable, reviewable scenario, and so gaps between Logic and Scenarios can be checked.

BLOCKING (ambiguity): Conditions 1 and 2 ('Add entry Toxicity Grade Date overlaps' / 'gaps') give no definition of what constitutes an overlap or a gap between consecutive Toxicity Grade date ranges (e.g., is a gap of 0 days allowed? Must stop date of one record equal onset date of the next, or onset date + 1?). → Define explicit rules, e.g. 'Record N+1 Onset Date must equal Record N Stop Date + 1 day; if less, it's an overlap, if greater, it's a gap' -- with example dates.

WARNING (ambiguity): Condition 5 error triggers when 'Add-entry section is blank... or because all the records are completed with UNK/ND/NA,' but the spec doesn't state whether this check happens per-AE or per-subject, nor how it interacts with the later rule that 'Add-entry records with both dates as UNK/ND/NA are not considered for scenarios 1 to 4' -- it's unclear if condition 5 fires only when ALL add-entry records are UNK/ND/NA, or if a mix of some valid + some UNK/ND/NA records could still partially trigger 5. → Clarify: 'Condition 5 applies only when every add-entry record for the AE has both Tox Grade Onset and Stop Date = UNK/ND/NA (or the add-entry section has zero records)'. State explicitly this is evaluated at the AE level.

WARNING (ambiguity): The exclusion rule 'Add-entry records with both dates as UNK/ND/NA are not to be considered for scenarios 1 to 4 when there are other records with data entered' does not clarify what 'other records with data entered' means -- does one valid date (e.g., only Onset filled) count as 'data entered', or must both dates be present? → Specify precisely, e.g. 'when at least one other add-entry record for the same AE has a non-UNK/ND/NA value in either Onset or Stop Date'.

WARNING (consistency): Conditions 4a, 4b, and 4c all output the identical error message 'AE End not latest Tox Grade Stop' despite representing three distinct situations (date mismatch, ongoing-but-no-blank-stop, dated-but-blank-stop). This makes it impossible to distinguish which condition fired from the output alone, which conflicts with the more descriptive, parameterized messages used in conditions 1 and 2 (which embed actual dates). → Either give 4a/4b/4c distinct messages describing the actual mismatch (as done for overlap/gap), or explicitly state that a single generic message is intentional for all three sub-cases.

WARNING (testability): Because 4a, 4b, and 4c produce the same Error Message text, and the spec lists no distinguishing Column Heading field (e.g., a scenario code) that separates these three sub-conditions, a Report Scenario built to test 4b (ongoing AE, no blank stop) cannot be distinguished from one testing 4c (dated AE, blank stop present) purely from the reported output. → Add a way to distinguish outputs for 4a/4b/4c in the Column Heading (e.g., an internal rule code) or accept in the spec that scenarios for these three sub-cases are verified only via input data, not output.

WARNING (completeness): The Logic never explicitly addresses how deleted individual Toxicity Grade add-entry records (as opposed to deleted subjects) should be treated in the gap/overlap/min/max calculations, even though 'Report should not display deleted records' is stated generally. → Clarify whether 'deleted records' refers only to deleted subjects/AEs or also individual deleted add-entry Tox Grade rows, and confirm deleted add-entry rows are excluded from min/max and gap/overlap calculations.

INFO (completeness): The Randomization filter states 'If Randomized only parameter selection checkbox is checked, the report should only output subjects with Randomization number that is not null' but doesn't specify the Form/Item name where Randomization Number is captured. → Reference the specific Form/Item (e.g., 'Randomization Number on the Randomization form') to align with naming conventions used elsewhere, even though this is a minor gap.

4. Generated Code

CQL — Guardrails: passed

```
SELECT ae.`@HDR`.`Site`.`Country` AS `Country`, ae.`@HDR`.`Site`.`Number` AS `Site`,
ae.`@HDR`.`Subject`.`Name` AS `Subject Number`, ae.AESEQ AS `AE Seq`, ae.AETERM AS `AE Term`,
ae.AESTDAT AS `AE Onset Date`, ae.AEENDAT AS `AE End Date`, cur.AGSTDAT AS `Tox Grade Onset Date`,
cur.AGENDAT AS `Tox Grade Stop Date`, CONCAT('Overlap with Tox Grade Onset ',
DATE_FORMAT(prev.AGSTDAT, '%d-%b-%Y'), ' and Tox Stop ', DATE_FORMAT(prev.AGENDAT, '%d-%b-%Y')) AS
`Error Message`
FROM `EDC`.`AEFM` ae
JOIN `EDC`.`AGTOXFM` cur ON cur.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND cur.AESEQ =
ae.AESEQ
JOIN `EDC`.`AGTOXFM` prev ON prev.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND prev.AESEQ =
ae.AESEQ AND prev.AGSTDAT < cur.AGSTDAT AND NOT EXISTS (SELECT 1
FROM `EDC`.`AGTOXFM` mid
WHERE mid.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND mid.AESEQ = ae.AESEQ AND mid.AGSTDAT
> prev.AGSTDAT AND mid.AGSTDAT < cur.AGSTDAT AND mid.`@Form`.`Status` = 'submitted__v' AND
(mid.DELETEDFM = 'N' OR mid.DELETEDFM IS NULL))
WHERE ae.`@Form`.`Status` = 'submitted__v' AND cur.`@Form`.`Status` = 'submitted__v' AND
prev.`@Form`.`Status` = 'submitted__v' AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND
(cur.DELETEDFM = 'N' OR cur.DELETEDFM IS NULL) AND (prev.DELETEDFM = 'N' OR prev.DELETEDFM IS NULL)
AND NOT (prev.AGSTDAT IS NULL AND prev.AGENDAT IS NULL) AND NOT (cur.AGSTDAT IS NULL AND
cur.AGENDAT IS NULL) AND prev.AGENDAT IS NOT NULL AND cur.AGSTDAT <= prev.AGENDAT AND
(`@PARAM`.`RandomizedOnly` = 0 OR EXISTS (SELECT 1
FROM `EDC`.`RANDFM` r
WHERE r.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND r.`@Form`.`Status` = 'submitted__v'
AND r.RANDNO IS NOT NULL))
UNION ALL
SELECT ae.`@HDR`.`Site`.`Country`, ae.`@HDR`.`Site`.`Number`, ae.`@HDR`.`Subject`.`Name`, ae.AESEQ,
ae.AETERM, ae.AESTDAT, ae.AEENDAT, cur.AGSTDAT, cur.AGENDAT, CONCAT('Gap with previous Tox Grade
Onset ', DATE_FORMAT(prev.AGSTDAT, '%d-%b-%Y'), ' and Tox Stop ', DATE_FORMAT(prev.AGENDAT, '%d-%b-
%Y'))
FROM `EDC`.`AEFM` ae
JOIN `EDC`.`AGTOXFM` cur ON cur.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND cur.AESEQ =
ae.AESEQ
JOIN `EDC`.`AGTOXFM` prev ON prev.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND prev.AESEQ =
ae.AESEQ AND prev.AGSTDAT < cur.AGSTDAT AND NOT EXISTS (SELECT 1
FROM `EDC`.`AGTOXFM` mid
WHERE mid.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND mid.AESEQ = ae.AESEQ AND mid.AGSTDAT
> prev.AGSTDAT AND mid.AGSTDAT < cur.AGSTDAT AND mid.`@Form`.`Status` = 'submitted__v' AND
(mid.DELETEDFM = 'N' OR mid.DELETEDFM IS NULL))
WHERE ae.`@Form`.`Status` = 'submitted__v' AND cur.`@Form`.`Status` = 'submitted__v' AND
prev.`@Form`.`Status` = 'submitted__v' AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND
(cur.DELETEDFM = 'N' OR cur.DELETEDFM IS NULL) AND (prev.DELETEDFM = 'N' OR prev.DELETEDFM IS NULL)
AND NOT (prev.AGSTDAT IS NULL AND prev.AGENDAT IS NULL) AND NOT (cur.AGSTDAT IS NULL AND
cur.AGENDAT IS NULL) AND prev.AGENDAT IS NOT NULL AND SUBDATE(cur.AGSTDAT, INTERVAL 1 DAY) >
prev.AGENDAT AND (`@PARAM`.`RandomizedOnly` = 0 OR EXISTS (SELECT 1
FROM `EDC`.`RANDFM` r
WHERE r.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND r.`@Form`.`Status` = 'submitted__v'
AND r.RANDNO IS NOT NULL))
UNION ALL
SELECT ae.`@HDR`.`Site`.`Country`, ae.`@HDR`.`Site`.`Number`, ae.`@HDR`.`Subject`.`Name`, ae.AESEQ,
ae.AETERM, ae.AESTDAT, ae.AEENDAT, tg.MinOnset, tg.MaxStop, 'AE Onset not earliest Tox Grade Onset'
FROM `EDC`.`AEFM` ae
JOIN (SELECT `@HDR`.`Subject`.`ID` AS SubjID, AESEQ AS AESEQ, MIN(AGSTDAT) AS MinOnset,
MAX(AGENDAT) AS MaxStop, SUM(CASE WHEN AGENDAT IS NULL THEN 1 ELSE 0 END) AS BlankStopCount
FROM `EDC`.`AGTOXFM`
WHERE `@Form`.`Status` = 'submitted__v' AND (DELETEDFM = 'N' OR DELETEDFM IS NULL) AND NOT (AGSTDAT
IS NULL AND AGENDAT IS NULL)
GROUP BY `@HDR`.`Subject`.`ID`, AESEQ) tg ON tg.SubjID = ae.`@HDR`.`Subject`.`ID` AND tg.AESEQ =
ae.AESEQ
WHERE ae.`@Form`.`Status` = 'submitted__v' AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND
RAWDATE(ae.AESTDAT) <> RAWDATE(tg.MinOnset) AND (`@PARAM`.`RandomizedOnly` = 0 OR EXISTS (SELECT 1
FROM `EDC`.`RANDFM` r
WHERE r.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND r.`@Form`.`Status` = 'submitted__v'
AND r.RANDNO IS NOT NULL))
UNION ALL
SELECT ae.`@HDR`.`Site`.`Country`, ae.`@HDR`.`Site`.`Number`, ae.`@HDR`.`Subject`.`Name`, ae.AESEQ,
ae.AETERM, ae.AESTDAT, ae.AEENDAT, tg.MinOnset, tg.MaxStop, 'AE End not latest Tox Grade Stop'
FROM `EDC`.`AEFM` ae
JOIN (SELECT `@HDR`.`Subject`.`ID` AS SubjID, AESEQ AS AESEQ, MIN(AGSTDAT) AS MinOnset,
MAX(AGENDAT) AS MaxStop, SUM(CASE WHEN AGENDAT IS NULL THEN 1 ELSE 0 END) AS BlankStopCount
```

```
FROM `EDC`.`AGTOXFM`
WHERE `@Form`.`Status` = 'submitted__v' AND (DELETEDFM = 'N' OR DELETEDFM IS NULL) AND NOT (AGSTDAT
IS NULL AND AGENDAT IS NULL)
GROUP BY `@HDR`.`Subject`.`ID`, AESEQ) tg ON tg.SubjID = ae.`@HDR`.`Subject`.`ID` AND tg.AESEQ =
ae.AESEQ
WHERE ae.`@Form`.`Status` = 'submitted__v' AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND
((ae.AEENDAT IS NOT NULL AND (tg.BlankStopCount > 0 OR RAWDATE(ae.AEENDAT) <> RAWDATE(tg.MaxStop)))
OR (ae.AEENDAT IS NULL AND tg.BlankStopCount = 0)) AND (`@PARAM`.`RandomizedOnly` = 0 OR EXISTS
(SELECT 1
FROM `EDC`.`RANDFM` r
WHERE r.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND r.`@Form`.`Status` = 'submitted__v'
AND r.RANDNO IS NOT NULL))
UNION ALL
SELECT ae.`@HDR`.`Site`.`Country`, ae.`@HDR`.`Site`.`Number`, ae.`@HDR`.`Subject`.`Name`, ae.AESEQ,
ae.AETERM, ae.AESTDAT, ae.AEENDAT, NULL, NULL, 'Add entry missing'
FROM `EDC`.`AEFM` ae
WHERE ae.`@Form`.`Status` = 'submitted__v' AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND
ae.AETERM IS NOT NULL AND NOT EXISTS (SELECT 1
FROM `EDC`.`AGTOXFM` t
WHERE t.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND t.AESEQ = ae.AESEQ AND
t.`@Form`.`Status` = 'submitted__v' AND (t.DELETEDFM = 'N' OR t.DELETEDFM IS NULL) AND NOT
(t.AGSTDAT IS NULL AND t.AGENDAT IS NULL)) AND (`@PARAM`.`RandomizedOnly` = 0 OR EXISTS (SELECT 1
FROM `EDC`.`RANDFM` r
WHERE r.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND r.`@Form`.`Status` = 'submitted__v'
AND r.RANDNO IS NOT NULL))
ORDER BY `Country`, `Site`, `Subject Number`, `AE Seq`
```

No study-metadata block was supplied, so Form/Item names are inferred: AE data comes from `EDC`.`AEFM` (AESEQ, AETERM, AESTDAT, AEENDAT), the toxicity-grade add-entry section is assumed to be a separate repeating Form `EDC`.`AGTOXFM` (AGSTDAT/AGENDAT) linked to the AE by Subject + AESEQ, and randomization number comes from an assumed `EDC`.`RANDFM`.RANDNO -- all three should be confirmed via DESCRIBE FORM before running, along with the deleted-flag item name (assumed DELETEDFM on each form). The five scenarios are implemented as UNION ALL branches: overlap/gap detection via a self-join with a NOT EXISTS gap-filler on AGTOXFM (SUBDATE used for the >1-day gap test), AE Onset/End mismatches via an aggregated MIN/MAX/blank-count subquery over non-fully-blank rows (which naturally implements the "exclude UNK/ND/NA rows only when other data-bearing rows exist" rule), and a final NOT EXISTS branch for "Add entry missing" when no data-bearing AGTOXFM row exists for a completed AETERM. dd-mon-yyyy formatting uses DATE_FORMAT(..., '%d-%b-%Y') and the Randomized-only checkbox is modeled as `@PARAM`.`RandomizedOnly` with an EXISTS check against RANDNO, consistent with the prior verified spec's pattern.